Rainfall Patterns and Short-Term Probabilistic Prediction in Singapore, 2017–2024

Course: DSA5208
Group members: [Names and student IDs]
Submission date: [Date]

Abstract

Briefly describe the dataset, the two project objectives, the use of Apache Spark, the main weather-pattern findings, and the predictive performance of the final model.

[Write this section after completing the experiments. Report only verified findings and results.]

1. Introduction

1.1 Background and Objectives

Introduce the historical five-minute rainfall measurements collected by monitoring stations across Singapore between 2017 and 2024.

State the two objectives:

- Investigate rainfall patterns within a year and differences across years.
- Estimate the probability that a specified station will accumulate more than a given rainfall threshold during the next 30 minutes.

1.2 Project Overview

Summarize the workflow from data validation and temporal preparation to exploratory analysis, feature construction, model development, and evaluation.

2. Data and Computing Environment

2.1 Dataset Description

Describe the data source, temporal resolution, geographical coverage, units, and key fields. Explain the distinction between observation timestamps and update timestamps.

Table 1. Dataset overview by year: observation counts, station counts, and temporal coverage.

2.2 Computing Environment

Report the actual environment used for each stage:

- Operating system and hardware, including CPU and available RAM.
- Python, Java, and Spark versions.
- Local or cluster execution mode.
- Spark worker threads, driver memory, and shuffle partitions.
- Data storage formats and relevant execution settings.

If group members used different environments, describe them separately.

2.3 Use of Apache Spark

Explain which processing steps use Spark DataFrames, aggregations, joins, and window functions, and which models and evaluation components use Spark MLlib.

Clarify that plotting operates on aggregated outputs rather than loading the complete observation dataset into pandas.

3. Data Quality Assessment and Preparation

3.1 Data Validation

Describe checks for missing values, duplicate station–timestamp keys, invalid rainfall values, inconsistent units, and station metadata changes.

Table 2. Data-quality checks, findings, and treatment decisions.

3.2 Temporal Alignment and Duplicate Resolution

Explain the historical timestamp conventions, the five-minute alignment rule, and how original timestamps were preserved.

Describe the 86 aligned-key collisions, the evidence used to assess them, and the deterministic record-selection rule.

Report observation counts before and after preparation.

3.3 Missing Observations and Coverage

Define coverage and explain how missing observations differ from zero rainfall.

Describe station entry and exit, extended gaps, and any suspected sensor issues. State which checks were diagnostic and which led to changes in the data.

Figure 1. Monthly station coverage and qualifying station counts.

3.4 Analysis-Specific Data Preparation

Explain the different requirements of the two tasks:

- Task 1: coverage-qualified aggregation and comparable rainfall summaries.
- Task 2: complete future target windows, historical feature availability, and treatment of incomplete histories.

Document any additional cleaning introduced by the prediction pipeline.

4. Task 1: Exploration of Rainfall Patterns

4.1 Analytical Approach

Define the rainfall measures, wet-period definitions, aggregation levels, coverage cutoffs, and station-weighting rules.

Explain exposure-normalized rainfall rates and distinguish them from observed totals. Describe how changing station participation is addressed.

4.2 Seasonal Patterns Within a Year

Discuss variation across months and the consistency of the seasonal pattern across years.

Figure 2. Monthly seasonal cycle.

4.3 Interannual Differences, 2017–2024

Compare years and identify unusually wet or dry periods after accounting for seasonal variation.

Discuss station participation and sensitivity to coverage thresholds.

Figure 3. Monthly anomalies and annual rainfall comparisons.

4.4 Diurnal Patterns

Describe how rainfall frequency and intensity vary through the day.

If included, examine whether the daily pattern changes by month or season.

Figure 4. Diurnal rainfall cycle.
Optional Figure 5. Month-by-hour rainfall patterns.

4.5 Spatial Differences

Compare rainfall across stations and regions using a clearly defined, temporally comparable measure.

Figure 5 or 6. Relative station rainfall map.

4.6 Rainfall Concentration and Extreme Observations

Explain the concentration of rainfall in a small fraction of time intervals and discuss the implications of the highly skewed distribution.

Figure 6 or 7. Rainfall concentration across temporal scales.

4.7 Implications for Prediction

Connect the descriptive findings to candidate time, station, recent-rainfall, and neighbouring-station features.

Distinguish exploratory evidence from predictive improvements that must be demonstrated on held-out data.

5. Task 2: Prediction Methodology

5.1 Prediction Target and Information Available at Forecast Time

Define the target probability:

$$
P\left(R_{s,(t,t+30\text{ min}]} > q \mid \mathcal{I}_t\right),
$$

where (s) is the station, (t) is the prediction time, (q) is the rainfall threshold, and ($\mathcal{I}_t$) contains information available at that time.

Explain the station, prediction-time, and threshold inputs, together with how historical observations are retrieved.

State assumptions about observation availability and reporting delays.

5.2 Label Construction and Probability Representation

Explain how the next six five-minute intervals form the future 30-minute total and how incomplete windows are handled.

Describe the chosen prediction approach:

- Direct threshold-specific binary classification; or
- Rainfall-bin classification followed by conversion to exceedance probabilities.

For the bin-based approach, specify the bin boundaries, rounding rules, treatment of thresholds within bins, and handling of the open-ended upper bin.

5.3 Feature Engineering

Describe each feature group, its motivation, calculation window, and availability at prediction time.

Table 3. Feature definitions and sources.

Potential groups include station attributes, periodic time features, recent local rainfall, and recent neighbouring-station rainfall. Include only features actually implemented.

5.4 Training, Validation, and Test Design

Document the chronological split:

- Training: 2017–2022.
- Validation: 2023.
- Testing: 2024.

Explain boundary handling, prevention of future-information leakage, training-only estimation of learned transformations, and any sampling and weighting strategy.

5.5 Baseline Models

Describe the implemented baselines:

- Global historical class or exceedance frequencies.
- Conditional climatology.
- Rainfall persistence or recent-state transition probabilities.

Explain how each baseline produces probabilities and how its parameters are estimated.

5.6 Machine Learning Models

Describe Logistic Regression and Random Forest, their Spark MLlib implementations, input features, and probability outputs.

Explain how hyperparameters are selected using the validation set.

Table 4. Model configurations and tuning ranges.

5.7 Evaluation Criteria and Experimental Comparisons

Define probability-quality metrics, such as Brier score, log loss, and calibration. Include PR-AUC for discrimination and RPS if ordinal rainfall bins are used.

Specify comparisons across rainfall thresholds, current wet/dry states, and other relevant subsets.

Use the same evaluation samples and feature sets when comparing algorithms. Describe feature-ablation experiments separately from algorithm comparisons.

6. Task 2: Results and Discussion

6.1 Baseline Performance

Present validation results and explain the strengths and weaknesses of the simple forecasts.

6.2 Model Selection and Final Test Performance

Explain the choice of the final configuration using validation results, then report performance on the held-out 2024 test set.

Table 5. Baseline and model performance by rainfall threshold.

6.3 Probability Calibration

Assess whether predicted probabilities agree with observed frequencies.

Figure 8. Calibration curves for selected thresholds.

6.4 Contribution of Feature Groups

Compare local features with local-plus-neighbour features while holding the model family fixed.

Table 6. Feature-ablation results.

6.5 Error Analysis

Discuss performance during dry-to-wet transitions, ongoing rainfall, heavy rainfall, and periods or stations with weaker predictions.

Support explanations with observed results rather than assuming that a more complex model must perform better.

7. Limitations and Further Work

Discuss limitations relevant to the implemented analysis, including:

- Incomplete observations and changing station coverage.
- Potentially informative missingness and uncertain sensor behaviour.

- Reporting delays and the distinction between retrospective and operational prediction.
- Limited years of data and temporal dependence.
- Rainfall-bin approximation and rare extreme events, where applicable.
- Missing meteorological variables and limits on causal interpretation.
- Computational constraints and useful extensions.

8. Conclusion

Answer both project objectives directly. State the main rainfall-pattern findings, the predictive value demonstrated by the experiments, and the principal limitations.

9. Declaration of AI Usage

Identify the AI tools used, the work they assisted with, and how the group reviewed and verified their outputs.

[Describe actual use in coding, debugging, analysis planning, writing, or editing. Do not claim checks that were not performed.]

References

List dataset sources, software documentation, and any meteorological or methodological references cited in the report.

Appendices

Appendix A. Detailed Data-Quality Results

Appendix B. Additional Task 1 Figures and Sensitivity Analyses

Appendix C. Full Feature Definitions and Hyperparameter Settings

Appendix D. Additional Evaluation Results

Appendix E. Reproduction Instructions and Submitted Model Files